

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

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Homework 26

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. Modify the `mpi_hello.c` program so that processes print their messages in rank order ($0 \rightarrow N-1$), instead of all at once. You may not use `sleep()`. Use MPI communication (such as `MPI_Send` and `MPI_Recv`) to enforce ordering. After implementing, run the program with multiple nodes and verify that the output always appears in strict rank order regardless of scheduling.
2. Write a program where each process generates a random integer. Use `MPI_Reduce` to compute the total sum of all values at rank 0. Print both the local values (per process) and the final sum.

- **Welcome: Sean**

- [LogOut](#)

